

Chongran Zhao

Ph.D. Student in Engineering · Brown University

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EDUCATION



Brown University

Ph.D. in Engineering

Mechanics of Solids and Structures

Advisor: Prof. Haneesh Kesari

Providence, RI

Sep. 2026 – Present



Southern University of Science and Technology

M.Eng. in Mechanics

B.S. in Theoretical and Applied Mechanics

Advisor: Prof. Ju Liu

Shenzhen, China

Sep. 2023 – Jun. 2026

Sep. 2019 – Jun. 2023

RESEARCH INTERESTS

Continuum and solid mechanics · Viscoelasticity · Constitutive modeling

Computational modeling · Numerical methods · Soft materials · Growth and remodeling

PUBLICATIONS

1. **A continuum and computational framework for viscoelastodynamics: III. A nonlinear theory**
J. Liu, J. Guan, **C. Zhao**, and J. Luo. *Computer Methods in Applied Mechanics and Engineering* 430 (2024), 117248.
2. **Modeling finite viscoelasticity based on the Green–Naghdi kinematic assumption and generalized strains**
J. Liu, **C. Zhao**, and J. Guan. *Journal of the Mechanics and Physics of Solids* 206 (2026), 106346.
3. **A framework for finite-strain viscoelasticity based on rheological representations**
C. Zhao, H. Yuan, and J. Liu. *International Journal of Solids and Structures* 330 (2026), 113893.

CONFERENCES & ACADEMIC ACTIVITIES

The Chinese Congress of Theoretical and Applied Mechanics

Changsha, China · Oral presentation

Jul. 18–21, 2025

A finite viscoelastic constitutive theory based on Green–Naghdi decomposition: A rheological model study

18th International Conference on Computational Plasticity

Barcelona, Spain · Oral presentation

Sep. 2–5, 2025

A rational framework of finite viscoelasticity: Dissipation potential and non-Newtonian effects

Summer School on Physics-Informed Modeling, Simulation and Experiments with Emphasis on the Cardiovascular System

Graz, Austria · Participant

Sep. 15–19, 2025

SKILLS

Technical C++ · CMake · MATLAB · Linux/Unix · Git · $\text{\LaTeX}$

Languages Chinese · English